

EE 3202.304

ECE Fundamentals Lab II

Lab 3: BJT - DC

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I. Experimental Results

Part 1

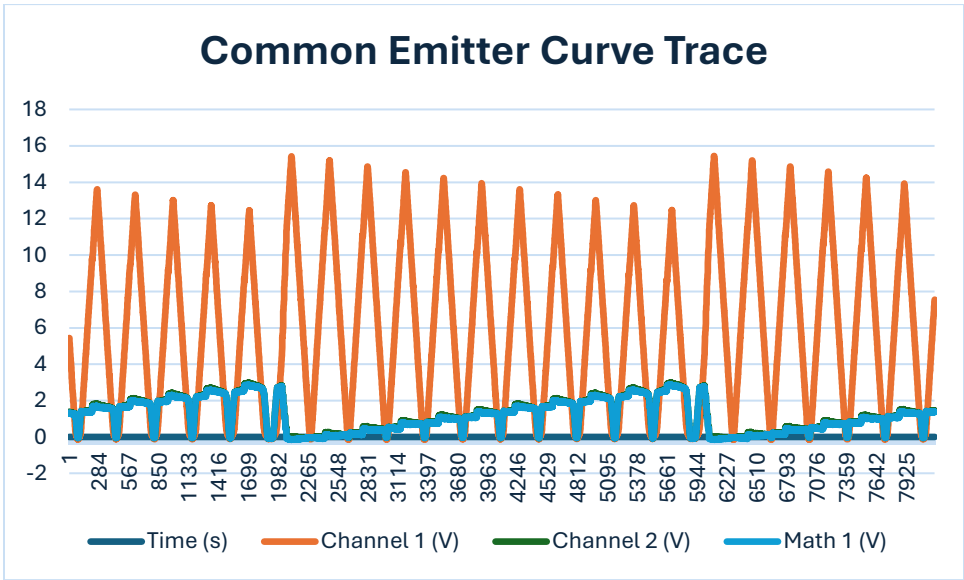


Figure 1.1: BJT Common Emitter Circuit IV Characteristics

BJT IV Curve Trace - Common Emitter

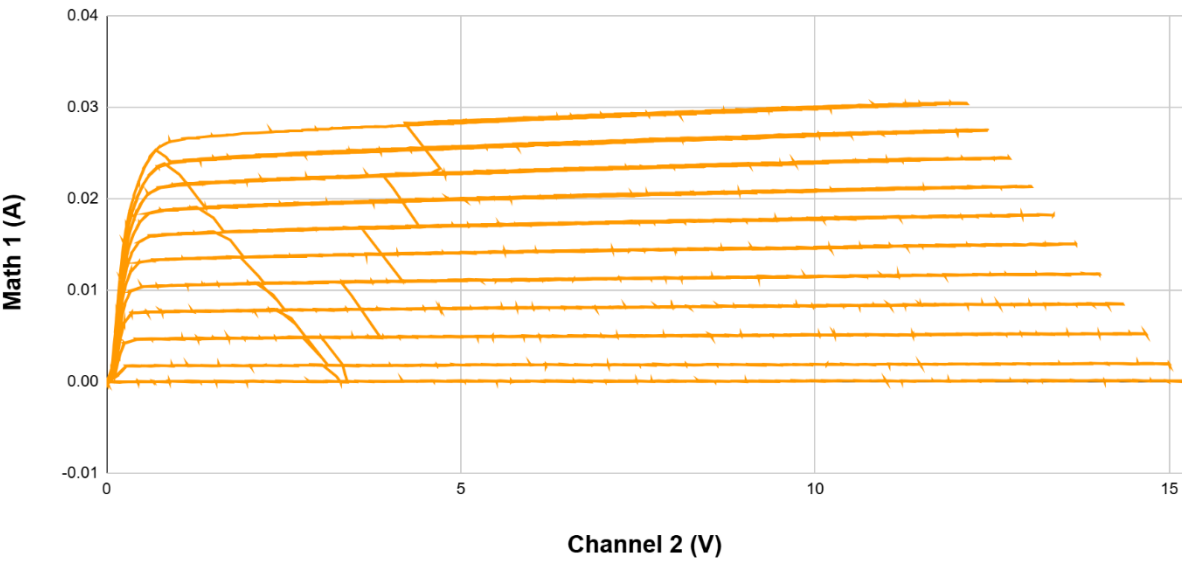


Figure 1.2: BJT Common Emitter Circuit IV Curve Trace

Part 2

vi	vbe	vce	vbc	BJT operating region
0	0.0184	5.0143	-4.9959	cut-off
0.1	0.1182	5.0143	-4.8961	cut-off
0.2	0.2192	5.0143	-4.7951	cut-off
0.3	0.3164	5.0143	-4.6979	cut-off
0.4	0.4117	5.0143	-4.6026	cut-off
0.5	0.4836	5.0088	-4.5252	cut-off
0.6	0.5228	4.9373	-4.4145	cut-off
0.7	0.6968	3.7506	-3.0538	active
0.8	0.7042	1.4297	-0.7255	active
0.9	0.7169	0.1976	0.5193	saturation
1	0.7173	0.1654	0.5519	saturation

Figure 2.1: BJT Inverting Switch Voltage Drops and Operating Region

Part 3

IV Curve Trace - Darlington Pair

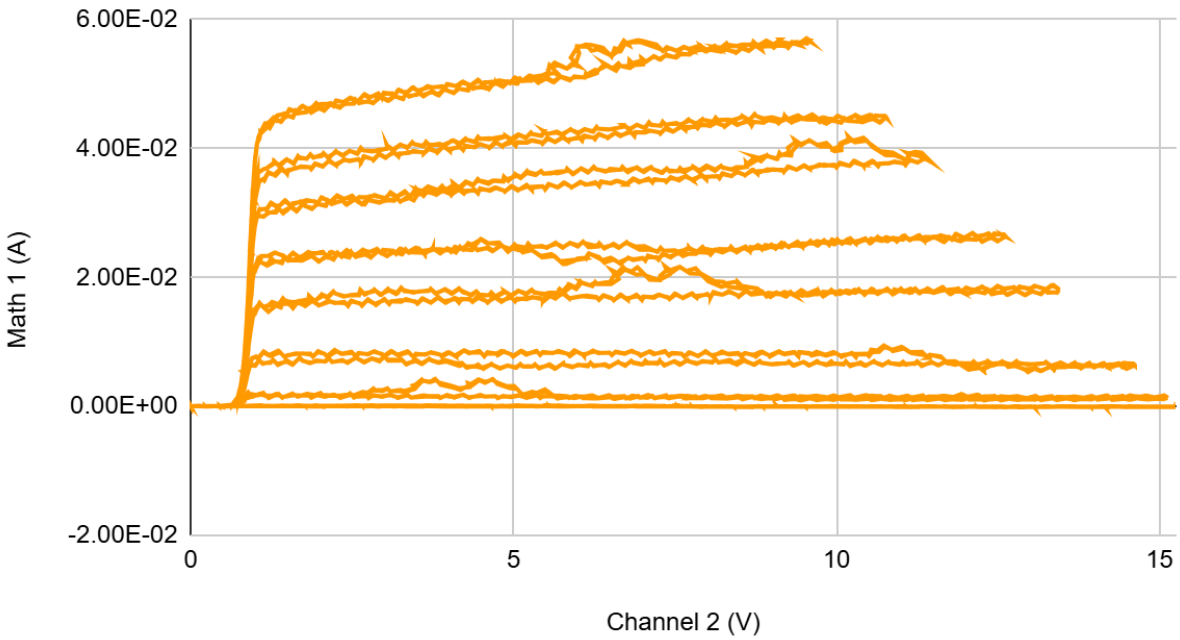


Figure 3.1: BJT Darlington Pair Circuit IV Curve Trace

II. Analysis and Thought

Part 1

The IV curve we obtained matches with the curve from the pre-lecture reading. Each curve showed a quick increase in collector current and then a flat line to show saturation. These curves are shown for each base current. In our curve shown in Figure 1.2, the curves were not perfect as there was noise interference, and the curves were not perfectly linear in the active region. These differences can be due to a multitude of factors like internal resistance and temperatures in our transistor. To limit the changes throughout the course of the lab we ensured to use the same transistor, so we can get the most accurate data possible.

The collector-emitter voltage V_{CE} , base current I_B , and collector current I_C were all close to what we expected, though not exactly the same. Using these values, we calculated the current gain, $\beta = I_C/I_B$, which was a little lower than the nominal value we assumed in the pre-lab. This small difference isn't surprising because β can vary between individual transistors even of the same type. Other factors like slight variations in resistor values, minor voltage drops in the breadboard connections, and small temperature changes during the experiment also probably contributed. Overall, the transistor behaved as expected and stayed in the active region, and our measured operating point was very close to the theoretical one from the pre-lab.

Looking at all our results together, the transistor behaved pretty much as we expected from theory and the datasheet. The base-emitter voltage during conduction was around 0.7 V, and the collector-emitter voltage in saturation was about 0.2 V, which matches the typical values for a 2N3904. Our measured β was also within the typical range listed in the datasheet, though slightly lower than the exact value we assumed in the pre-lab. Any small differences between theory and our measurements are normal and can be explained by real-world factors like internal resistance, leakage currents, slight temperature changes, and small variations between individual transistors. Overall, the transistor showed the expected cutoff, active, and saturation behavior, and our measurements confirmed that it was operating correctly throughout the experiment.

Part 2

Yes, the BJT successfully acted like a switch. As the input increased, we observed the corresponding changes in V_{CE} , V_B , and V_E as seen in Figure 2.1. The transistor's three main operating states were cutoff, active, and saturated. At low input voltages, it was in the cutoff, with no measurable collector current and V_{CE} near the supply voltage. As the input reached around 0.7 V, the transistor entered the active region, and the collector current increased proportionally with the base current. When the input voltage was high enough, the transistor reached saturation, with V_{CE} dropping to approximately 0.2 V and the collector was fully on. These observations confirm that the BJT can function effectively as a switch, transitioning cleanly between the states.

Part 3

The Darlington pair is basically two BJTs connected so that the current amplified by the first transistor is fed into the second, giving a much higher overall current gain than a single transistor. In our experiment, we built the Darlington pair following the pre-lab design and looked at how it behaved. Compared to a regular BJT amplifier, the IV curve looks similar overall, but there are a few noticeable differences. The base-emitter voltage is higher roughly the sum of the two junctions, so it turns on at a slightly higher input voltage. Also, the collector's current rises much more quickly with small changes in base current, showing a much higher gain. This setup makes the Darlington pair useful for driving larger loads while still showing the usual cutoff, active, and saturation regions like a normal BJT.

III. References

None